THz-Net: A Multi-Scale Self-Attention Residual Network for Cross-System Terahertz Image Reconstruction

RUNMIN JIANG,[1,2] CHENG TANG,[2] JIE GAO,[1,2] JIYUE JIANG, [3] SIQI WU,[4] ZHIYONG TAN,[1,2] AND JUNCHENG CAO[1,2,*]

[1]*State Key Laboratory of Materials for Integrated Circuits, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai 200050, China*
[2]*Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China*
[3]*Department of Computer Science and Engineering, The Chinese University of Hong Kong, Sha Tin 999077, New Territories, Hong Kong SAR, China*
[4]*School of Computer Science and Information Security, Guilin University of Electronic Technology, Guilin 541010, Guangxi, China*
[*]*jccao@mail.sim.ac.cn*

Abstract: High-resolution reconstruction of terahertz (THz) images is critical for a wide range of imaging applications, such as biomedical diagnosis and non-destructive testing. However, THz images inherently suffer from diffraction-induced blur, low contrast, and loss of fine structural detail. Existing reconstruction methods are limited by insufficient high-frequency feature extraction, lack of multi-scale feature integration, and poor generalization across imaging systems with different optical configurations and noise characteristics. To address these limitations, we propose THz-Net, a multi-scale self-attention residual network for cross-system THz image reconstruction. The network employs an adaptive residual backbone to preserve subtle contrast variations, a multi-scale feature extraction module to capture complementary structural information at different network depths, and a dual-attention module that sequentially applies channel and spatial attention to selectively enhance informative features while suppressing background interference. We evaluate THz-Net on images acquired from three THz transmission-mode imaging systems with distinct optical configurations. Experimental results show that THz-Net achieves an average PSNR of 37.41 dB and SSIM of 0.97, outperforming the best baseline by 3.93 dB in PSNR and 0.08 in SSIM, and maintains consistent reconstruction quality across all three systems.

1. Introduction

Terahertz (THz) radiation occupies the electromagnetic spectrum between the infrared and microwave regions, spanning frequencies from 0.1 to 10 THz. Its relatively short wavelength, substantial penetration depth in non-metallic materials, non-ionizing photon energy, and capacity to convey rich spectral signatures have made THz imaging attractive for a growing range of applications, including biomedical diagnosis [1–4], non-destructive testing [5, 6], security screening [7, 8], agricultural sensing [9, 10], and industrial quality control [11–13]. However, the relatively long wavelengths of THz radiation, pronounced diffraction effects, and inherent system noise collectively degrade the spatial resolution and edge fidelity of acquired images [14, 15]. THz images typically exhibit diffraction-induced boundary blur, low contrast, ringing-like artifacts, and loss of fine structural detail, all of which complicate downstream analysis and limit the utility of THz imaging in precision applications [16]. Enhancing the resolution of THz images through computational reconstruction is therefore of considerable practical importance.

Two general strategies have been pursued to improve THz image quality: system-level hardware upgrades and algorithm-based reconstruction [17]. System-level approaches can improve optical resolution but often incur high cost and significant complexity, and the achievable improvements

remain constrained by the diffraction limit [18]. Algorithmic methods, by contrast, seek to recover high-resolution detail directly from degraded images acquired by standard THz imaging systems, offering a flexible and cost-effective alternative [19, 20]. Early numerical approaches, including Lucy–Richardson deconvolution [18, 21], projection onto convex sets [22, 23], and iterative back-projection [22], demonstrated the feasibility of computational resolution enhancement but remained limited by their dependence on explicit degradation modeling and extensive parameter tuning. More recently, deep learning methods have substantially advanced THz image reconstruction. Convolutional neural networks tailored for THz degradation patterns were shown to outperform classical algorithms in feature extraction and detail recovery [24, 25], while adaptive feature modification layers [26] and complex-domain CNN extensions [14] broadened the scope of learnable representations. Residual channel attention mechanisms were introduced to strengthen high-frequency feature capture [16], and encoder-decoder architectures such as U-Net variants demonstrated effectiveness in jointly addressing blur, noise, and downsampling artifacts [15, 27]. Further contributions have explored time-domain prior embedding [28], deep learning-based computed tomography [29, 30], and efficient architectures with optimized training strategies [31].

Despite this progress, several challenges remain that are closely tied to the distinctive degradation characteristics of THz images. THz images suffer from spatially diffuse intensity distributions and weak edge contrast caused by diffraction [32], and standard convolutional layers with limited receptive fields struggle to distinguish meaningful structural boundaries from background fluctuations under such conditions, often producing reconstructions with blurred edges and incomplete texture recovery [33]. Moreover, different feature channels in THz images carry unequal amounts of useful information: some channels encode structural content while others are dominated by noise or low-contrast background. Without an explicit mechanism to selectively emphasize informative channels and suppress uninformative ones, networks tend to treat all features equally, diluting the contribution of critical high-frequency components [34, 35]. The structural information relevant to THz image reconstruction also spans multiple spatial scales, from fine-grained textures captured at shallow network layers to coarse global contours encoded at deeper layers; single-depth feature extraction cannot simultaneously capture these complementary representations [36]. A further limitation is that existing approaches are typically developed and evaluated under a single imaging configuration, yet THz imaging systems in practice vary considerably in beam formation, detector characteristics, point spread function profiles, and noise levels. Models trained on one system frequently produce over-smoothed results or boundary artifacts when applied to images from a different system, and this lack of cross-system robustness remains a significant barrier to practical deployment [37].

To address these challenges, we propose THz-Net, a multi-scale self-attention residual network designed for cross-system THz image reconstruction. The network employs an adaptive residual backbone that removes batch normalization and post-addition activation to preserve the subtle contrast and intensity variations that are particularly important in THz imagery, where useful structural signals often reside in low-intensity regions. A multi-scale feature extraction module collects feature maps from multiple backbone stages, enabling the network to jointly exploit fine-grained detail from shallow layers and broader contextual patterns from deeper layers. A dual-attention module is then applied independently at each scale: the channel attention branch aggregates global spatial statistics through pooling to selectively weight feature channels according to their informativeness, while the spatial attention branch highlights structurally salient regions and suppresses background interference. The refined multi-scale features are subsequently aligned and fused through a post-processing head to produce the final high-resolution output. By combining targeted feature selection with hierarchical feature integration, THz-Net recovers sharp edges and fine textures from degraded THz inputs while maintaining robustness across imaging systems with distinct degradation characteristics.

In this paper, we construct three THz transmission-mode imaging systems with different optical configurations and combine the acquired images into a unified cross-system test set. THz-Net is compared against several representative baseline methods under the same data and degradation settings, and reconstruction quality is evaluated using both quantitative metrics and visual comparisons. Ablation studies confirm that the channel attention, spatial attention, and multi-scale feature fusion modules each contribute meaningfully to the overall reconstruction performance. Experimental results demonstrate that THz-Net achieves consistent improvements across all three imaging systems, confirming its effectiveness for cross-system high-resolution THz image reconstruction.

2. Methods

2.1. Preliminaries: THz Image Formation

Terahertz (THz) imaging exploits electromagnetic waves to probe material properties through characteristic transmission and spectral responses. However, the relatively long wavelength of THz radiation causes the acquired images to suffer from diffraction-limited blur, ringing-like boundary artifacts, and loss of fine structural detail. In practice, absorption and attenuation in the sample further reduce signal strength and image contrast, making high-resolution THz image reconstruction a challenging inverse problem.

The image formation process can be modeled as a convolution of the underlying scene with the system point spread function (PSF), followed by system-dependent sampling and noise corruption:

$$y = \mathcal{S}\big(x * \mathbf{h}\big) + \mathbf{z}, \tag{1}$$

where x denotes the underlying high-resolution image, y is the observed degraded measurement, $\mathbf{h}$ is the effective PSF, $*$ denotes two-dimensional convolution, $\mathcal{S}(\cdot)$ represents the sampling and discretization operator, and $\mathbf{z} \sim \mathcal{N}(0, \beta^2)$ is additive Gaussian noise with standard deviation β. For analytical simplicity, the effective PSF is approximated by a Gaussian kernel:

$$\mathbf{h}(u, v) \approx G_\rho(u, v) = \frac{1}{2\pi\rho^2} \exp\Big(- \frac{u^2 + v^2}{2\rho^2} \Big), \tag{2}$$

where (u, v) are kernel coordinates and ρ controls the effective beam width.

Under this degradation model, THz image reconstruction aims to recover a high-resolution estimate $\hat{x}$ from y:

$$\hat{x} = g(y; \Phi), \tag{3}$$

where $g(\cdot; \Phi)$ denotes the reconstruction network parameterized by Φ. Because y is simultaneously affected by blur, sampling loss, and noise, accurate reconstruction demands both detail restoration and robustness to system-dependent degradation.

In practice, the forward model varies across different THz imaging systems. The PSF $\mathbf{h}$ may differ in width, shape, and spatial invariance; the sampling operator $\mathcal{S}$ changes with detector pitch, array resolution, and acquisition geometry; and the noise level β depends on source power, detector characteristics, and system design. Mismatch between the training and test-time degradation conditions can therefore cause ringing, over-smoothing, contrast loss, or artifact amplification. To address these challenges, we propose THz-Net, a multi-scale self-attention residual network for cross-system high-resolution THz image reconstruction.

2.2. THz-Net Framework

2.2.1. Overview

As illustrated in Fig. 1, THz-Net reconstructs a high-resolution image through four coordinated stages: (1) an *adaptive residual backbone* (Part 1) that extracts hierarchical features from the

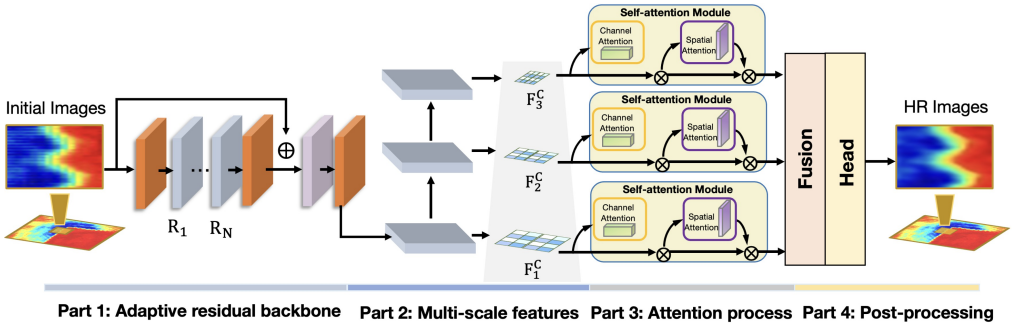


Fig. 1. Overall architecture of THz-Net. The framework consists of four main parts: an adaptive residual backbone (Part 1), multi-scale feature extraction (Part 2), a self-attention processing module (Part 3), and a post-processing fusion head for image reconstruction (Part 4).

degraded input; (2) a *multi-scale feature extraction module* (Part 2) that collects complementary representations at different network depths; (3) a *dual-attention module* (Part 3) that enhances informative responses while suppressing background artifacts; and (4) a *post-processing fusion head* (Part 4) that aggregates the refined multi-scale features and produces the final high-resolution output. By integrating these components, THz-Net provides a unified reconstruction framework capable of handling system-dependent degradations across diverse THz imaging platforms.

2.2.2. Adaptive Residual Backbone

THz-Net adopts a modified residual backbone for hierarchical feature extraction. Compared with the original ResNet [38], batch normalization layers are removed to avoid constraining feature intensity variation, which helps preserve subtle contrast and texture details in THz images, especially in low-intensity regions. The ReLU activation after the residual addition is also removed to better maintain local contrast during reconstruction. A comparison between the original and the proposed adaptive residual block is shown in Fig. 2(a).

Let $\mathbf{F}_0$ denote the shallow feature extracted from the input image y by an initial convolution layer. The backbone consists of N stacked adaptive residual blocks $\mathcal{R}_1, \ldots, \mathcal{R}_N$:

$$\mathbf{F}_n = \mathcal{R}_n(\mathbf{F}_{n-1}), \qquad n = 1, 2, \ldots, N, \tag{4}$$

where $\mathbf{F}_n$ is the output feature map of the n-th block. As shown in Fig. 2(a), each adaptive residual block takes an input X_l and produces

$$X_{l+1} = X_l + \alpha\, \mathcal{T}(X_l), \tag{5}$$

where $\mathcal{T}(\cdot)$ denotes the residual branch (Conv–ReLU–Conv) and α is a constant scaling factor. In our implementation, $\alpha = 0.1$ is used for wide backbones and is omitted when the channel width is moderate (e.g., 64 channels per layer). A global residual connection is applied at the end of the backbone, adding the initial shallow feature $\mathbf{F}_0$ to the final block output $\mathbf{F}_N$ (denoted by $\oplus$ in Fig. 1).

2.2.3. Multi-Scale Feature Extraction

THz images commonly exhibit diffraction-induced stripes, low contrast, and blurred boundaries, making fine structural recovery difficult with features from a single network depth. To capture information at multiple receptive fields and abstraction levels, the multi-scale feature extraction module collects output feature maps from the last three stages of the backbone and denotes them

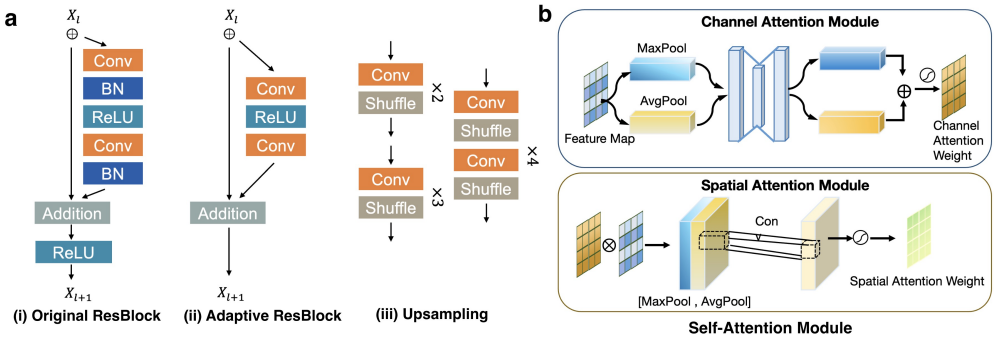


Fig. 2. Detailed schematics of the key network components. (a) Comparison between the original residual block and the proposed adaptive residual block, together with the upsampling unit. (b) Structure of the dual-attention module, which sequentially applies channel attention and spatial attention.

as $\mathbf{F}_1^{\mathrm{C}}$, $\mathbf{F}_2^{\mathrm{C}}$, and $\mathbf{F}_3^{\mathrm{C}}$ (see Part 2 of Fig. 1), where $\mathbf{F}_i^{\mathrm{C}} \in \mathbb{R}^{C_i \times H_i \times W_i}$. These feature maps encode complementary structural information at different depths and are forwarded in parallel to the subsequent dual-attention module (Part 3) for further refinement.

2.2.4. Dual-Attention Module

To enhance salient features and suppress background interference, a dual-attention module $\mathcal{D}(\cdot)$ is applied independently to each of the three multi-scale feature maps $\{\mathbf{F}_i^{\mathrm{C}}\}_{i=1}^{3}$, as shown in Part 3 of Fig. 1 and detailed in Fig. 2(b). The module sequentially applies channel attention and spatial attention. For notational clarity, we describe the operation on a generic input feature map $\mathbf{F} \in \mathbb{R}^{\bar{C} \times \bar{H} \times \bar{W}}$ (representing any $\mathbf{F}_i^{\mathrm{C}}$). The refined output is computed as

$$\mathbf{F}' = \mathbf{M}_{\mathrm{ch}}(\mathbf{F}) \otimes \mathbf{F}, \qquad \mathbf{F}'' = \mathbf{M}_{\mathrm{sp}}(\mathbf{F}') \otimes \mathbf{F}', \tag{6}$$

where $\mathbf{M}_{\mathrm{ch}} \in \mathbb{R}^{\bar{C} \times 1 \times 1}$ and $\mathbf{M}_{\mathrm{sp}} \in \mathbb{R}^{1 \times \bar{H} \times \bar{W}}$ are the channel and spatial attention maps, respectively, and $\otimes$ denotes element-wise multiplication with broadcasting.

Channel attention. Global spatial information is aggregated via average pooling and max pooling to produce two channel descriptors $\mathbf{F}_{\mathrm{avg}}^{\mathrm{ch}}$, $\mathbf{F}_{\mathrm{max}}^{\mathrm{ch}} \in \mathbb{R}^{\bar{C} \times 1 \times 1}$. These descriptors are each fed through a shared two-layer MLP and the results are summed and activated:

$$\mathbf{M}_{\mathrm{ch}}(\mathbf{F}) = \sigma\Big(\mathbf{W}_1\big(\mathbf{W}_0\,\mathbf{F}_{\mathrm{avg}}^{\mathrm{ch}}\big) + \mathbf{W}_1\big(\mathbf{W}_0\,\mathbf{F}_{\mathrm{max}}^{\mathrm{ch}}\big)\Big), \tag{7}$$

where $\sigma(\cdot)$ is the sigmoid function, and $\mathbf{W}_0$, $\mathbf{W}_1$ are the shared MLP weights with a reduction ratio r in the hidden layer.

Spatial attention. Channel-wise average and max pooling are applied to obtain two spatial descriptors $\mathbf{F}_{\mathrm{avg}}^{\mathrm{sp}}$, $\mathbf{F}_{\mathrm{max}}^{\mathrm{sp}} \in \mathbb{R}^{1 \times \bar{H} \times \bar{W}}$. The two maps are concatenated along the channel dimension and processed by a 7×7 convolution:

$$\mathbf{M}_{\mathrm{sp}}(\mathbf{F}) = \sigma\Big(f^{7\times 7}\big([\mathbf{F}_{\mathrm{avg}}^{\mathrm{sp}};\, \mathbf{F}_{\mathrm{max}}^{\mathrm{sp}}]\big)\Big), \tag{8}$$

where $[\cdot\,;\cdot]$ denotes channel-wise concatenation. Applying the above dual-attention operation to each multi-scale feature map yields the refined representations:

$$\hat{\mathbf{F}}_i^{\mathrm{C}} = \mathcal{D}(\mathbf{F}_i^{\mathrm{C}}), \qquad i = 1, 2, 3. \tag{9}$$

The three refined feature maps $\hat{\mathbf{F}}_1^{\mathrm{C}}$, $\hat{\mathbf{F}}_2^{\mathrm{C}}$, and $\hat{\mathbf{F}}_3^{\mathrm{C}}$ are subsequently aligned to a common resolution, concatenated, and fused through a convolutional layer in the post-processing head (Part 4), which produces the final high-resolution reconstruction $\hat{x}$.

3. Experiments

3.1. Cross-System THz Imaging Setups

To evaluate the cross-system generalization capability of THz-Net, three THz imaging systems with distinct optical configurations were employed, as shown in Fig. 3. Although all three systems operate in transmission mode at 4.3 THz, they differ in beam formation, illumination extent, detector characteristics, and resulting image contrast. These differences introduce realistic domain shifts and provide a suitable testbed for assessing the robustness and transferability of THz-Net across heterogeneous THz imaging conditions.

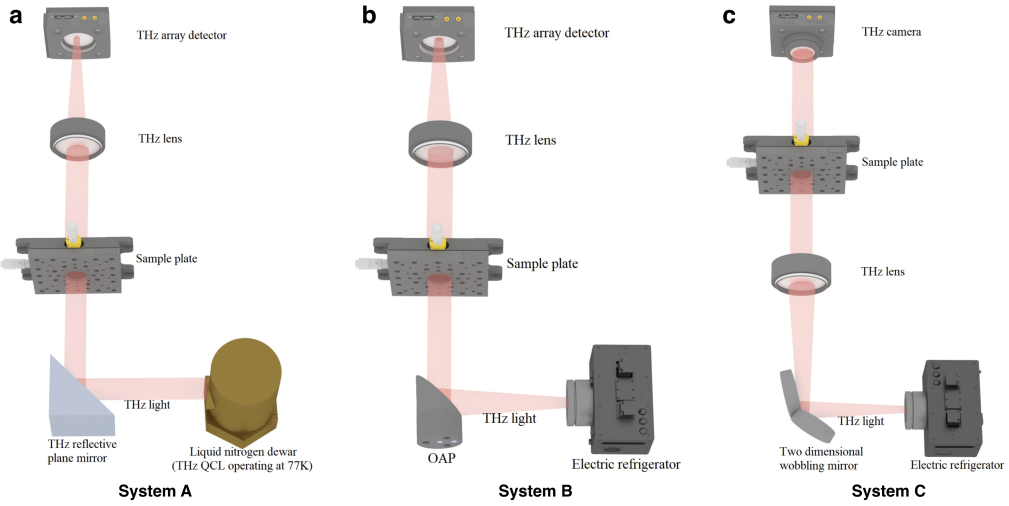


Fig. 3. Terahertz Imaging Optical Systems

System A: Quasi-parallel single-lens imaging. As shown in Fig. 3(a), this setup uses a collimated 4.3 THz quantum cascade laser (QCL) in transmission mode. A single anti-reflection-coated high-resistivity silicon plano-convex lens (38.1 mm diameter, 50 mm focal length, transmittance > 90%) forms the image. With a beam diameter of approximately 5 mm, the system provides concentrated illumination and high image contrast. The detector is a VO_x-based focal plane array (GAT, FPA-P1740; 384 × 288 pixels, 17 μm pixel pitch, 25 Hz). The THz power at the sample plane is approximately 1 mW.

System B: Externally collimated single-lens imaging. As shown in Fig. 3(b), this configuration also operates at 4.3 THz, but uses an externally collimated beam expanded to approximately 15 mm, making it suitable for medium-sized samples. The same silicon lens and VO_x detector (FPA-P1740) are used. Compared with System A, this setup yields more moderate image contrast and slightly higher noise. The THz power at the sample plane is approximately 2 mW.

System C: Large-beam imaging with a two-dimensional scanning mirror. As shown in Fig. 3(c), this setup employs a divergent beam and a compound variable-focus lens group (NEC, IRV-TL028; 28.2 mm focal length, F/1.0, transmittance > 85%). A two-dimensional

galvanometric mirror rapidly scans the beam along both axes, generating a cluster of scanning beams that integrate into a nearly uniform illumination pattern at the detector plane. The detector is a VO_x-based focal plane array (IRV-T0831C; 320×240 pixels, 23.5 μm pixel pitch, 8.5 Hz). This configuration was designed for imaging biological cancer tissue samples, which exhibit strong absorption, low signal-to-noise ratio, and weak contrast, thereby representing the most challenging domain for cross-system generalization.

In summary, Systems A–C span progressively larger illumination areas and lower contrast levels, corresponding to compact-sample, medium-scale, and biological tissue imaging scenarios, respectively. The detailed optical and acquisition parameters of the three systems are summarized in Table 1. Together, these configurations capture realistic variability in THz imaging and define three distinct imaging domains for evaluating the generalization performance of THz-Net.

Table 1. Detailed optical and acquisition parameters of the three THz imaging systems.

Parameter	System A	System B	System C
Mode	Transmission	Transmission	Transmission
Frequency	4.3 THz	4.3 THz	4.3 THz
Wavelength	70 μm	70 μm	70 μm
Pixels	384×288	384×288	320×240
Pixel dimension	17 μm	17 μm	23.5 μm
Ambient temperature	$26 \pm 1\,^{\circ}$C	$25 \pm 1\,^{\circ}$C	$25 \pm 1\,^{\circ}$C
Humidity	$61\% \pm 2\%$	$45\% \pm 1\%$	$50\% \pm 2\%$

3.2. Network Training and Baseline Methods

Dataset and training configuration. We utilized the DIV2K dataset [39], which comprises 900 high-definition images, for model training (800 for training, 100 for validation and testing). Since paired high-resolution THz images are difficult to obtain in practice, we adopt a simulation-based training strategy: all training images are preprocessed using the degradation model described in Section 2.1, which applies the system PSF convolution, downsampling, and additive noise to synthesize degraded inputs that emulate realistic THz imaging conditions. This allows the network to learn the inverse mapping from degraded to clean images in a supervised manner, without requiring paired THz ground truth. For evaluation, THz images acquired from the three imaging systems described in Section 3.1 were combined into a unified cross-system test set. The network was trained using the Adam optimizer with Mean Squared Error (MSE) as the loss function. Since the human visual system is more sensitive to luminance than chrominance, reconstructed images were converted to the YCbCr color space, and both PSNR and SSIM were computed on the Y (luminance) channel for quantitative evaluation. All experiments were conducted on a single NVIDIA RTX 3090 Ti GPU.

Baseline methods. To comprehensively evaluate THz-Net, we compare it against several representative super-resolution methods. Bicubic interpolation [40] is included as a traditional non-learning baseline. For learning-based baselines, we select CNN-based methods spanning both plain convolutional models and residual variants. Specifically, SRCNN [25] and CRNN [41] are included as representative CNN models, while VDSR [42], EDSR [43], and ResNet [38]

represent deeper residual-learning architectures. We also include U-Net [15], an encoder-decoder network specifically designed for THz image high-rolution reconstruction. All baseline models were trained and evaluated under the same data and degradation settings for a fair comparison.

3.3. Comparison with State-of-the-Art Methods

Quantitative comparison. The quantitative results, averaged across the three THz imaging systems, are summarized in Table 2. THz-Net achieves the highest PSNR (37.41 dB) and SSIM (0.97) among all compared methods. Compared with the best-performing baseline EDSR (33.48 dB / 0.89), THz-Net obtains a gain of 3.93 dB in PSNR and 0.09 in SSIM. The traditional Bicubic interpolation yields only 30.68 dB / 0.81, confirming its inability to recover fine structures from degraded THz images. Early CNN-based methods such as SRCNN and VDSR provide marginal improvements over Bicubic, while more advanced architectures including EDSR and U-Net offer notably better performance but still fall short of THz-Net. These results indicate that the proposed multi-scale dual-attention design is particularly effective for THz image reconstruction, where simultaneously restoring fine details and suppressing system-dependent artifacts is critical.

Table 2. Quantitative comparison of THz-Net with state-of-the-art super-resolution methods on THz image reconstruction, averaged across three imaging systems. Best results are in **bold** and second-best are underlined. Absolute gains over the second-best are shown in parentheses.

Method	Bicubic	SRCNN	VDSR	EDSR	CRNN	ResNet	U-Net	**THz-Net**
PSNR (dB)	30.68	30.78	31.28	33.48	30.70	31.04	31.67	**37.41** (+3.93)
SSIM	0.81	0.82	0.85	0.89	0.81	0.83	0.86	**0.97** (+0.08)

Visual comparison. Fig. 4(a) presents a visual comparison on a THz image of a chip's peripheral contour. The Bicubic result suffers from noticeable color shift, blurring, and reduced brightness. Deep learning methods improve upon the traditional approach but still exhibit residual artifacts and inaccurate fine details in the highlighted regions. In contrast, THz-Net produces sharper edges, richer textures, and more faithful structural recovery.

Line-profile analysis. To quantitatively validate the visual improvements, we analyze the 1D pixel intensity profiles along the designated scanlines 'b' and 'c' in Fig. 4(b) and (c). The full-line profiles confirm that THz-Net successfully tracks the global structural trends of the original signal. The zoomed-in regions further reveal the limitations of existing methods: baseline networks tend to over-smooth local features, failing to recover the steep intensity gradients at structural boundaries. In contrast, THz-Net reconstructs sharper transitions that closely follow the ground truth, corresponding to improved structural fidelity. Moreover, in highly variable regions such as the valley in Fig. 4(b) and the peak area in Fig. 4(c), baseline methods struggle to maintain the dynamic range, resulting in flattened features or elevated noise floors. In contrast, THz-Net effectively suppresses irrelevant background interference while preserving critical high-frequency variations, yielding sharper localized contrast and better detail reconstruction. This capability is particularly important for accurate defect detection in industrial THz NDE applications.

3.4. Cross-System Generalization

Having established the superiority of THz-Net over existing methods, we further evaluate its generalization capability across different hardware configurations. As described in Section 3.1,

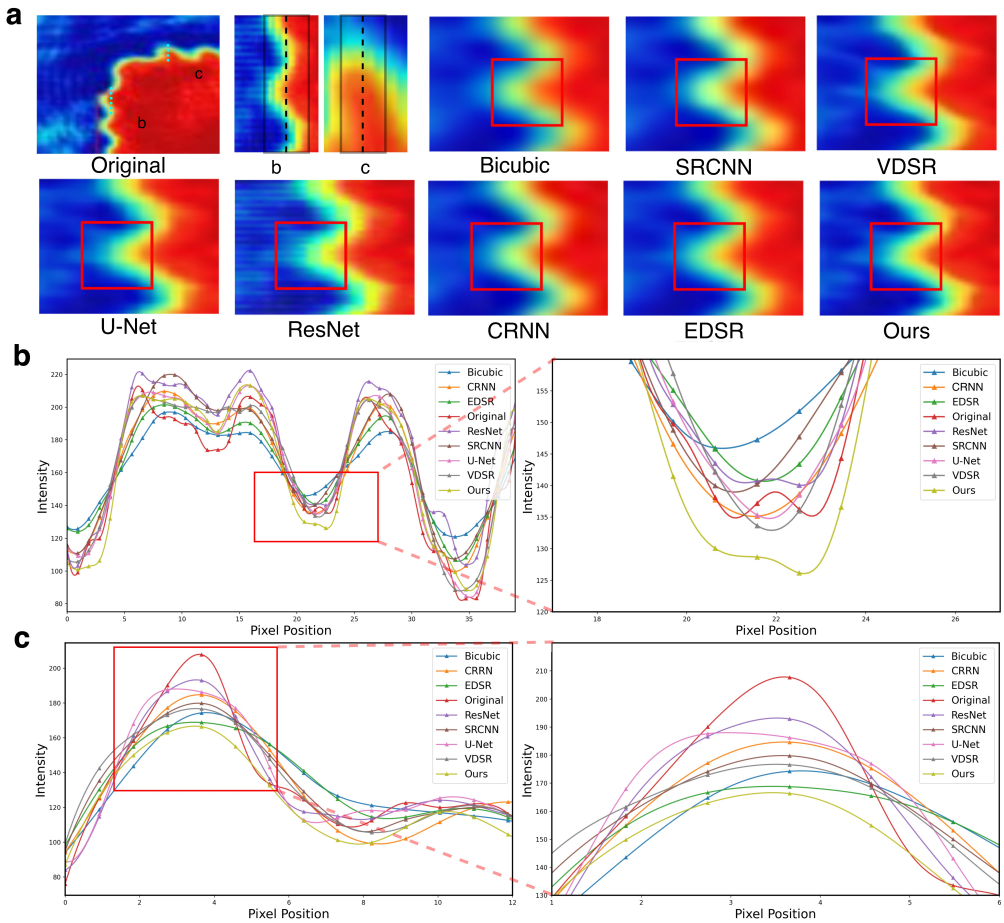


Fig. 4. Qualitative and quantitative comparison with SOTA methods. (a) Visual comparison on a Terahertz image. Dashed lines 'b' and 'c' on the ground truth (Original) indicate the cross-sections for the 1D profiles below. Red boxes highlight regions of interest for detail comparison. (b)–(c) 1D intensity profiles extracted along scanlines 'b' and 'c', respectively. The right plots show zoomed-in views of the red-boxed segments from the left plots.

the three imaging systems introduce distinct degradation characteristics, posing a challenging cross-domain reconstruction problem.

Domain characterization. Fig. 5 presents representative samples from each system, with the top row showing original 2D intensity images and the bottom row displaying the corresponding 3D intensity surfaces. System A exhibits relatively concentrated energy and clear structural boundaries; System B shows increased background fluctuation and reduced contrast; and System C presents the most challenging characteristics, with broad illumination, complex texture, and weak effective contrast. These visual differences directly reflect the distinct optical configurations and noise properties of each system.

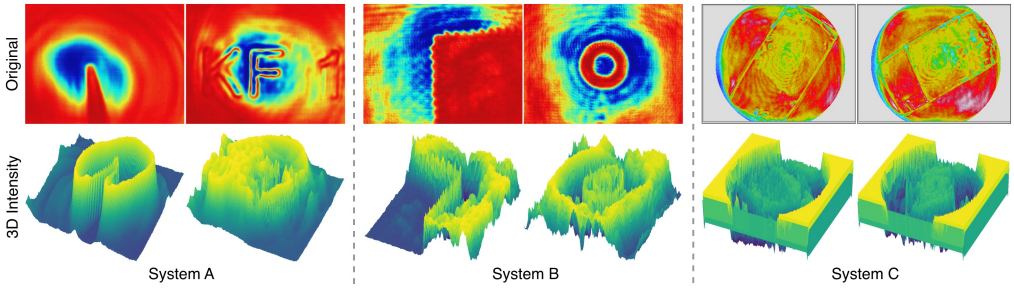


Fig. 5. Visualization of representative THz samples acquired by different imaging systems. The top row shows original 2D intensity images, and the bottom row presents the corresponding 3D intensity surfaces, where height denotes pixel intensity. The examples from Systems A–C reveal noticeable differences in contrast, spatial intensity pattern, and noise level across imaging domains.

To further quantify the inter-system distributional discrepancy, we visualize the dataset using t-SNE and UMAP, as shown in Fig. 6 (a) and (b), respectively. Both embeddings reveal a clear domain-dependent structure: System C forms a compact and well-separated cluster, indicating a pronounced distribution shift relative to the other two systems, while Systems A and B occupy neighboring regions with limited partial overlap, suggesting shared but distinguishable imaging characteristics. These consistent separation patterns confirm that the constructed test set covers multiple heterogeneous imaging domains, providing a realistic and challenging benchmark for evaluating cross-system robustness.

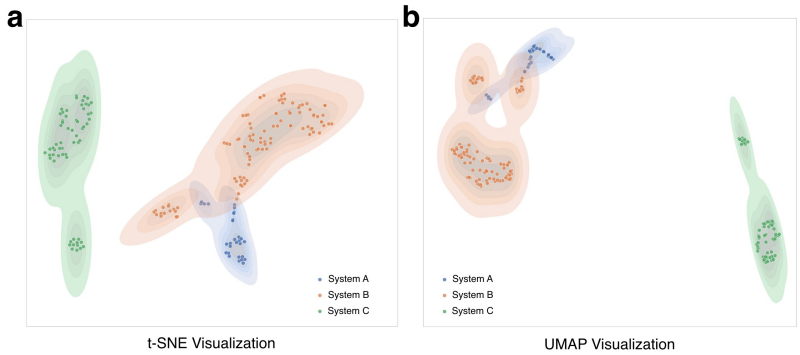


Fig. 6. Distribution visualization of the constructed THz dataset using (a) t-SNE and (b) UMAP. Samples from the three imaging systems form distinct but partially related clusters, revealing clear inter-system distribution discrepancies.

Quantitative cross-system performance. Despite the significant domain gaps among the three systems, THz-Net consistently achieves high PSNR and SSIM values across all domains without any system-specific fine-tuning, as summarized in Fig. 7. Specifically, the model attains PSNR values of 37.35 dB, 38.62 dB, and 36.26 dB on Systems A, B, and C, respectively, with corresponding SSIM values of 0.97, 0.98, and 0.96. Even on System C, which exhibits the most severe domain shift as confirmed by the t-SNE/UMAP analysis, the model retains strong reconstruction quality (36.26 dB / 0.96). These results demonstrate the robust self-adaptive capability of THz-Net across diverse hardware configurations, validating its practical applicability in real-world cross-system THz imaging scenarios.

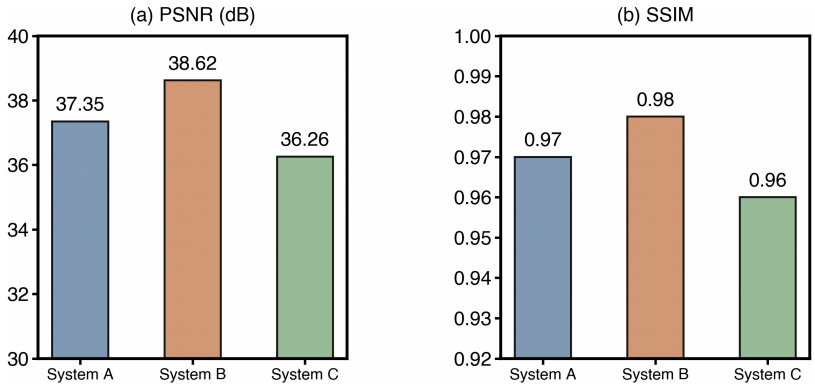


Fig. 7. Quantitative performance evaluation across different hardware configurations. Comparison of PSNR and SSIM metrics achieved by the proposed method on three distinct Terahertz imaging systems (System A, System B, and System C).

3.5. Ablation Study

Qualitative ablation. To evaluate the contribution of each key module, we conducted ablation experiments on two representative THz test images: a cancer tissue slide and a laser facet sample. Four configurations were compared: the full THz-Net model, and three variants with the channel attention module removed (w/o CAM), the spatial attention module removed (w/o SAM), and the multi-scale feature fusion module removed (w/o MSF), alongside the original ground truth for reference. As shown in Fig. 8 and Fig. 9, the full model achieves the sharpest structural edges and the richest fine textures in both samples. Removing CAM reduces local contrast and introduces edge artifacts; removing SAM weakens spatial coherence and leads to uneven detail distribution; and excluding MSF results in noticeable blurring and loss of fine-grained detail.

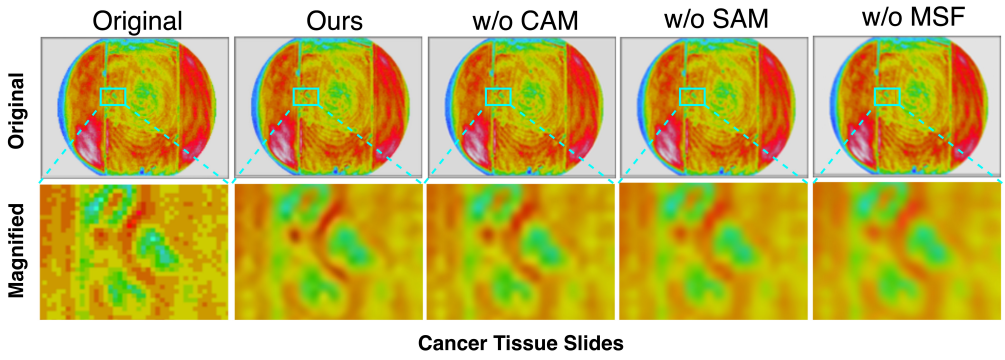


Fig. 8. Visual ablation study on cancer tissue slides. The figure compares the reconstruction results of the proposed model with configurations in which specific components were removed: without the Channel Attention Module (w/o CAM), without the Spatial Attention Module (w/o SAM), and without Multi-Scale Features (w/o MSF).

Quantitative ablation. The quantitative results in Table 3 corroborate the visual observations. Removing any single module leads to a substantial performance drop: the w/o SAM variant incurs the largest PSNR degradation (from 37.41 dB to 34.54 dB, a loss of 2.87 dB), indicating that spatial attention plays a critical role in preserving structural consistency. The w/o MSF variant

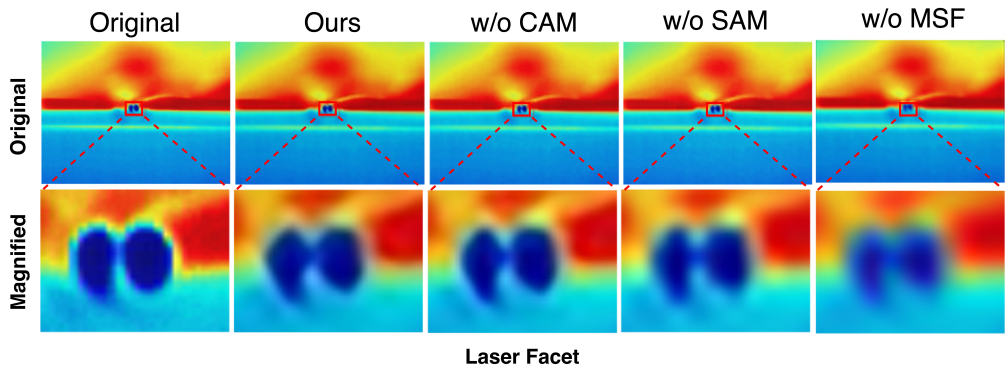


Fig. 9. Visual ablation study on Laser Facet samples. Evaluation of the contribution of different network modules to the reconstruction of industrial laser facet images.

suffers the most significant SSIM reduction (from 0.97 to 0.92), confirming that multi-scale feature fusion is essential for maintaining perceptual quality and fine-grained structural similarity. The w/o CAM variant also exhibits a notable decline (PSNR dropping to 34.88 dB), demonstrating that channel attention contributes meaningfully to feature selection and contrast enhancement. Together, these results confirm that all three modules are complementary and jointly essential for achieving optimal reconstruction quality.

Table 3. Ablation study on key network modules. Each variant removes one component while retaining the others. Best results are in **bold** and second-best are underlined. Absolute gains over the second-best are shown in parentheses.

Metric	w/o CAM	w/o SAM	w/o MSF	**THz-Net (Full)**
PSNR (dB)	34.88	34.54	34.68	**37.41** (+2.53)
SSIM	0.94	0.94	0.92	**0.97** (+0.03)

Generalization across diverse object structures and scales. To further assess the model's adaptability, we evaluated THz-Net on three representative sample types spanning industrial, biological, and metallic domains, with reconstruction performed at ×2, ×3, and ×4 upscaling factors. As illustrated in Fig. 10, the model achieves high-fidelity reconstruction at ×2, retaining fine textural details and producing sharp, artifact-free contours comparable to the ground truth. At higher magnification factors (×3 and ×4), minor high-frequency attenuation is observed—a natural consequence of the increasingly ill-posed nature of large-scale super-resolution—while the global structural integrity remains well preserved. These results confirm that THz-Net maintains stable performance across varying upscaling factors and generalizes effectively to diverse THz imaging scenarios.

Hyperparameter study. To determine the optimal configuration for THz-Net, we conducted a systematic hyperparameter search over the number of training epochs, learning rate, residual scaling factor, batch size, input patch size, and the number of residual blocks. The results are detailed in Table 4. The final model adopts the configuration that yields the best balance between reconstruction performance and training efficiency: 20 epochs, a learning rate of 5×10^{-6}, residual scaling of 0.05, batch size of 32, patch size of 192, and 32 residual blocks.

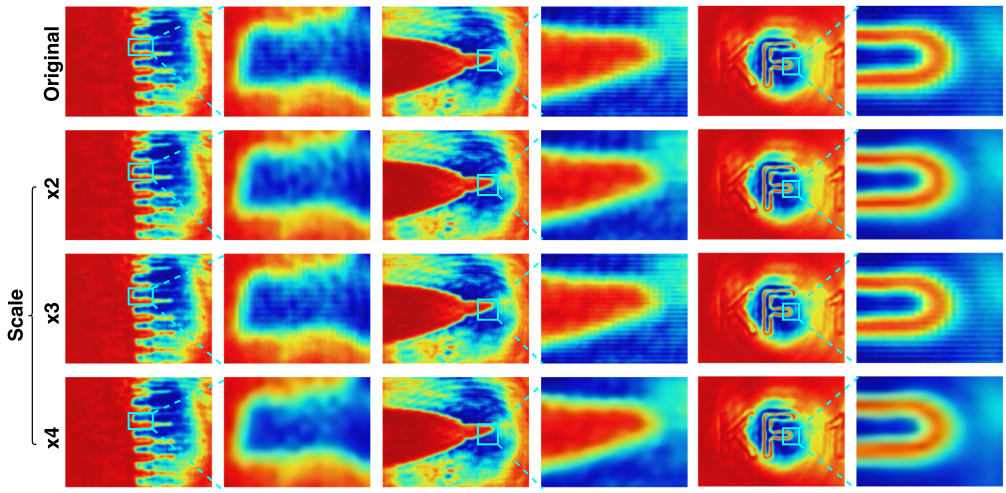


Fig. 10. Super-resolution reconstruction results across different scale factors. Visual comparison of Terahertz images reconstructed at ×2, ×3, and ×4 upscaling factors.

Table 4. Ablation studies on hyperparameter configurations. We evaluate the impact of different settings on model performance using PSNR (dB) and SSIM. The optimal configurations (marked in **bold**) are selected for our final model.

(a) Number of Epochs

Epochs	PSNR	SSIM
5	33.24	0.9420
10	37.41	0.9722
20	**38.37**	**0.9786**
50	37.95	0.9795

(b) Learning Rate

LR	PSNR	SSIM
1e-4	34.86	0.9635
1e-5	38.07	0.9789
1e-6	37.41	0.9722
5e-6	**38.56**	**0.9793**

(c) Residual Scaling

Scale	PSNR	SSIM
0.01	35.65	0.9496
0.05	**37.41**	**0.9722**
0.10	35.65	0.9565
0.15	34.86	0.9505

(d) Batch Size

Batch	PSNR	SSIM
8	35.68	0.9560
16	36.04	0.9591
32	**37.41**	**0.9722**
64	36.71	0.9648

(e) Patch Size

Patch	PSNR	SSIM
64	36.35	0.9610
128	36.47	0.9624
192	**37.41**	**0.9722**
256	36.39	0.9621

(f) Number of ResBlocks

Blocks	PSNR	SSIM
8	36.21	0.9567
16	35.76	0.9553
32	**37.41**	**0.9722**
64	36.03	0.9652

4. Conclusion

In this work, we addressed the problem of high-resolution THz image reconstruction under cross-system imaging variability, where diffraction-induced blur, weak contrast, and system-dependent degradation jointly limit the recovery of fine structural details. To this end, we proposed THz-Net, a multi-scale self-attention residual network that combines an adaptive residual backbone, hierarchical multi-scale feature extraction, and sequential channel-spatial attention to enhance informative structures while suppressing background interference. By integrating residual learning with attention-guided feature refinement across multiple depths, the proposed framework is able to recover both local detail and global structural consistency from degraded THz observations.

Experimental results on three THz transmission-mode imaging systems with distinct optical configurations demonstrated that THz-Net consistently outperformed representative baseline methods in both quantitative and qualitative evaluations, while maintaining strong robustness across heterogeneous imaging domains without system-specific fine-tuning. These results indicate that the proposed method provides an effective and practical solution for cross-system THz image reconstruction, with potential value for high-resolution THz imaging tasks in biomedical inspection, non-destructive testing, and other application settings where image quality and domain robustness are both critical. Future work will focus on extending the framework to broader real-world THz acquisition conditions and further improving robustness under more complex degradation patterns.

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